

NGO Management System Using Blockchain

Adpula Maruthi Prathap

Department of Computer Science and Engineering (Cyber Security)

CVR College of Engineering, Hyderabad, India

SK .Shahezeen Sulthana

Department of Computer Science and Engineering (Cyber Security)

CVR College of Engineering, Hyderabad, India

Dilli Sindhu

Department of Computer Science and Engineering (Cyber Security)

CVR College of Engineering, Hyderabad, India

ABSTRACT—Non-governmental organizations (NGOs) depend on public donations to conduct social-welfare activities, but donors often have limited visibility into how funds are collected, allocated, and spent. Conventional donation-management platforms typically rely on centralized databases and administrative reporting, which can create information asymmetry and make independent verification of transaction history difficult. This paper proposes an NGO Management Using Blockchain platform that combines a web application, role-based access control, conventional data storage, decentralized document storage, payment services, and blockchain-based transaction recording. The proposed architecture provides separate interfaces for administrators, NGOs, and donors, while managing users, campaigns, donations, expenditures, documents, transactions, and reports. MongoDB is used for core application records, IPFS is used for documents and receipts, and Solidity-based smart contracts on an Ethereum-compatible blockchain maintain immutable transaction records. Payment gateways such as Razorpay and Stripe, notification services, and an AI assistant are integrated as external services. The platform further includes an audit and verification layer that cross-checks information across the application, storage, and blockchain layers. The proposed system is designed to improve transparency, traceability, accountability, and donor confidence while reducing dependence on a single centralized record.

INDEX TERMS—Blockchain, NGO Management, Donation Transparency, Smart Contracts, Ethereum, IPFS, Auditability, Role-Based Access Control

I. INTRODUCTION

Non-governmental organizations play an important role in mobilizing financial and material resources for social welfare. A donation lifecycle generally involves identifying a campaign, receiving donations, recording transactions, allocating resources, and reporting expenditures. When these activities are managed through conventional centralized systems, donors may have to depend on organizational reports to understand how their contributions were used. This creates information asymmetry between donors, NGOs, administrators, and other stakeholders.

Blockchain technology provides a distributed ledger in which transactions can be recorded in an ordered and tamper-resistant manner. Its characteristics of traceability, transparency, cryptographic integrity, and programmable smart contracts make it suitable for applications where multiple parties need to verify transaction history. Prior research has specifically examined blockchain-based charity systems and has highlighted its potential to improve transparency, information sharing, fund allocation, and traceability in donation workflows [1], [7], [8], [11], [12].

The proposed work develops an NGO management platform in which blockchain is not used as a replacement for every application database. Instead, the architecture assigns different responsibilities to appropriate layers. MongoDB stores core application records, IPFS stores documents and receipts, and the blockchain layer records smart-contract transactions and immutable ledger information. The backend coordinates these

layers and applies authentication, role-based access control (RBAC), and logging.

The main objective of the proposed system is to provide a transparent and auditable environment for NGO operations. The platform supports three principal user categories—administrator, NGO, and donor—and manages users, campaigns, donations, expenditures, documents, transactions, and reports. External payment gateways and notification services are connected to the backend, while an AI assistant can support user interaction. The architecture supplied with the project explicitly identifies these layers and their cross-checking through an audit and verification component [13].

II. LITERATURE REVIEW

Blockchain-based approaches to charity and donation management have been studied as a means of reducing trust problems and improving traceability. Wu and Zhu proposed a blockchain-supported charity donation service system focused on information release, donation management, fund and material allocation, project management, and organizational administration [1]. Their work emphasizes that transparency and traceability can help address information asymmetry in charitable activities.

Zheng et al. provided an overview of blockchain architecture, consensus mechanisms, and application areas, establishing the technical basis for considering blockchain as a distributed ledger for transaction-oriented systems [7]. Lee et al. proposed a blockchain-based donation environment with an emphasis on transparency and privacy, demonstrating the relevance of blockchain to sustainable donation processes [8]. These studies

support the use of blockchain as a verifiable record layer rather than relying exclusively on centralized records.

Research has also examined blockchain in social emergency relief and charity applications. Jia and Deng discussed blockchain applications in social emergency relief, while Wang and Chen analyzed blockchain technology in Chinese charity [11], [12]. Qin et al. investigated the design and implementation of a charity system based on blockchain [13]. Nor et al. proposed a blockchain-based Sadaqa mechanism for disaster-aid crowdfunding, and Jayasinghe et al. studied philanthropy on the blockchain [14], [15]. Collectively, these works show that blockchain can support donation traceability and trust among stakeholders.

The proposed project differs in emphasis by integrating NGO operational management with the donation ledger. Rather than focusing only on a donation transaction, the architecture connects users, campaigns, donations, expenditures, documents, transactions, and reports with authentication, storage, payment, notification, and audit services. The use of MongoDB for operational records and IPFS for documents keeps large or frequently queried application data outside the blockchain, while the smart-contract layer provides an immutable record for blockchain-relevant transactions. This layered design is intended to balance usability, storage efficiency, and auditability.

III. METHODOLOGY

The proposed methodology follows a layered web-and-blockchain architecture. The workflow begins when an administrator, NGO representative, or donor accesses the React.js and Tailwind CSS frontend. Requests are sent to the Node.js/Express backend, where authentication, authorization, and logging are applied. JWT-based authentication establishes the user session, while RBAC limits operations according to the user's role.

A. User and Access Management: The administrator manages platform-level operations, while NGO users manage campaigns, donation-related records, expenditures, supporting documents, and reports. Donors interact with campaigns, make donations through the supported payment flow, and access relevant transaction and reporting information. The role separation reduces unauthorized access and establishes accountability for actions performed through the platform.

B. Campaign and Donation Management: NGOs create and manage campaigns through the application. Campaign information is stored as core application data. Donor actions generate donation records that can be associated with a campaign and a blockchain transaction. The application maintains transaction metadata for querying and reporting, while the blockchain provides an immutable reference for blockchain-recorded events.

C. Expenditure and Document Management: Expenditures are recorded against the relevant NGO or campaign. Supporting documents and receipts are stored through IPFS rather than placing large files directly on-chain. The application maintains references to these documents so that the evidence associated with an expenditure can be retrieved during auditing. This follows the general principle of separating operational data and document storage from the immutable transaction layer.

D. Blockchain and Smart Contract Layer: The blockchain component uses Solidity smart contracts and an Ethereum-compatible ledger. Smart contracts define the operations that are permitted to be recorded on-chain. Transaction information is represented in an immutable chain ledger, enabling later verification of the recorded event. The design follows the blockchain characteristics of sequential records, cryptographic integrity, and traceability discussed in prior research [1], [7].

E. External Services: Payment gateways such as Razorpay and Stripe provide the payment interface. Email and SMS services can deliver notifications related to campaigns and transactions. The architecture also identifies an AI assistant using OpenAI or Gemini as an optional service for user support. These services are connected through the backend rather than directly exposing the blockchain layer to end users.

F. Audit and Verification: The final layer cross-checks information from the application, storage, and blockchain components. A donation or expenditure can therefore be inspected using its application record, related documents, and blockchain transaction information. The audit layer is intended to make inconsistencies easier to detect and to provide a traceable basis for reporting.

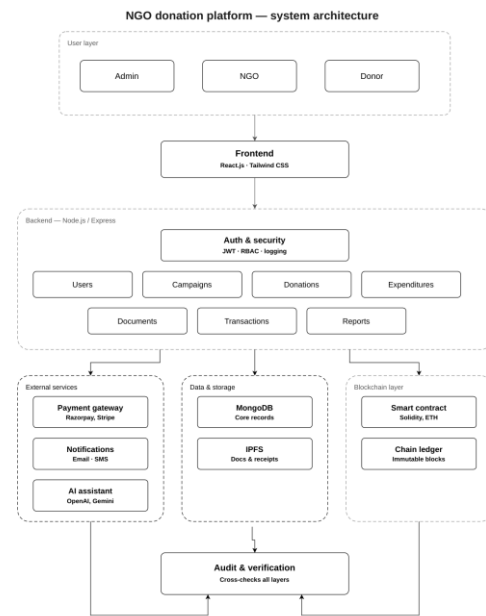


Fig. 1. Proposed NGO donation platform system architecture.

IV. SYSTEM WORKFLOW

The operational sequence of the proposed platform is as follows:

- 1) User registration and authentication are handled by the web application using JWT and RBAC.
- 2) An NGO creates a campaign and records campaign details in the application database.
- 3) A donor selects a campaign and initiates a donation through the supported payment gateway.

- 4) The backend records the donation metadata and invokes the appropriate smart-contract operation for the blockchain record.
- 5) The blockchain returns a transaction reference that can be associated with the application record.
- 6) When funds are used, the NGO records an expenditure and uploads supporting documents or receipts.
- 7) Documents are stored through IPFS and their references are associated with the expenditure record.
- 8) Reports combine application records with blockchain references and supporting documents.
- 9) The audit and verification layer cross-checks the available records and supports transparent reporting.

This workflow reflects the project architecture, in which the backend connects the user-facing system to MongoDB, IPFS, external services, and the blockchain layer [13].

V. SECURITY AND ACCOUNTABILITY

Security and accountability are addressed through multiple controls rather than through blockchain alone. JWT provides authenticated sessions, RBAC restricts operations according to role, and backend logging creates an application-level activity trail. Smart contracts provide controlled blockchain operations, while the immutable ledger helps preserve the integrity of blockchain-recorded transactions. IPFS provides decentralized addressing for documents and receipts, but access to sensitive documents should still be controlled by the application.

The architecture therefore uses defense in depth. Authentication protects entry to the platform; authorization controls actions; application records support normal queries; IPFS stores evidence documents; the blockchain records selected transactions; and the audit layer compares these sources. This layered approach is important because immutability does not by itself guarantee that the original information submitted to a system is truthful. Verification procedures and appropriate administrative controls remain necessary.

VI. ARCHITECTURAL OUTCOMES

The supplied project material defines a complete system architecture rather than presenting measured experimental results. Therefore, this paper reports architectural and functional outcomes instead of claiming numerical performance improvements. The resulting design provides:

- A three-role user layer consisting of administrator, NGO, and donor.
- A React.js/Tailwind CSS frontend connected to a Node.js/Express backend.
- JWT authentication, RBAC, and logging for application security.
- Dedicated management of users, campaigns, donations, expenditures, documents, transactions, and reports.
- MongoDB for core application records and IPFS for documents and receipts.
- Solidity smart contracts and an Ethereum-compatible chain ledger for immutable blockchain records.
- Integration points for Razorpay and Stripe payment gateways and email/SMS notifications.
- An optional AI assistant layer using OpenAI or Gemini.
- An audit and verification layer that cross-checks the application, document, and blockchain layers.

These components collectively address the central project objective: making donation-related records more transparent and easier to verify. The architecture also avoids storing all application data directly on-chain, which is consistent with the use of blockchain as a transaction and integrity layer rather than as a general-purpose file database.

VII. DISCUSSION

The proposed architecture offers several advantages. First, blockchain-backed transaction records can make post-transaction modification more difficult and can provide a common verification point for stakeholders. Second, separating operational data from blockchain data reduces the amount of information that must be written to the ledger. Third, IPFS provides a dedicated location for supporting documents and receipts, while MongoDB supports conventional application queries. Fourth, RBAC makes it possible to distinguish administrative, NGO, and donor operations.

However, several limitations must be considered. Blockchain transactions can introduce latency and transaction costs, and the scalability of the ledger must be evaluated before deployment at large scale. Smart contracts must be carefully tested because errors in contract logic can affect recorded transactions. IPFS addresses document integrity and distribution, but privacy and authorization still require application-level controls. Payment gateway integration introduces dependency on external services. Finally, blockchain immutability does not guarantee that a submitted expenditure or receipt is genuine; therefore, organizational verification and audit procedures remain necessary.

The current project materials do not provide benchmark measurements such as transaction latency, throughput, gas consumption, storage overhead, or user-acceptance scores. These metrics should be evaluated in future experimental work. A meaningful evaluation can compare a centralized-only implementation with the proposed layered implementation using measures such as record-verification time, transaction processing time, storage cost, and audit completeness.

VIII. CONCLUSION

This paper presented a proposed NGO Management Using Blockchain platform for improving transparency and auditability in donation management. The system combines a React.js/Tailwind CSS frontend, Node.js/Express backend, JWT and RBAC security, MongoDB application storage, IPFS document storage, Solidity smart contracts, and an Ethereum-compatible blockchain ledger. Payment gateways, notifications, and an optional AI assistant extend the platform's functionality.

The main contribution of the architecture is the integration of NGO operational records with an immutable blockchain transaction layer and a dedicated audit and verification component. Instead of placing every record on the blockchain, the design assigns application records, documents, and blockchain transactions to appropriate layers and connects them through verifiable references. This layered approach is intended to improve transparency without making the blockchain responsible for all application storage.

Future work should implement and benchmark the prototype, evaluate smart-contract security, measure transaction and storage overhead, strengthen privacy controls for documents, and conduct usability testing with NGO administrators and donors. With these evaluations, the proposed architecture can be developed into a practical platform for transparent and accountable NGO management.

REFERENCES

- [1] H. Wu and X. Zhu, “Developing a Reliable Service System of Charity Donation During the Covid-19 Outbreak,” *IEEE Access*, vol. 8, 2020, pp. 154848–154858.
- [2] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.
- [3] N. Z. Aitzhan and D. Svetinovic, “Security and privacy in decentralized energy trading through multi-signatures, blockchain and anonymous messaging streams,” *IEEE Trans. Dependable Secure Comput.*, vol. 15, no. 5, pp. 840–852, Sep. 2018.
- [4] G. Zheng, S. Xie, H. Dai, X. Chen, and H. Wang, “An overview of blockchain technology: Architecture, consensus, and future trends,” in *Proc. IEEE Int. Congr. Big Data (BigData Congress)*, 2017, pp. 557–564.
- [5] J. Lee, A. Seo, Y. Kim, and J. Jeong, “Blockchain-based one-off address system to guarantee transparency and privacy for a sustainable donation environment,” *Sustainability*, vol. 10, no. 12, p. 4422, Nov. 2018.
- [6] G. Peters and E. Panayi, “Understanding modern banking ledgers through blockchain technologies: Future of transaction processing and smart contracts on the Internet of money,” in *Banking Beyond Banks and Money*. Springer, 2016, pp. 239–278.
- [7] H. Jia and X. Deng, “A preliminary analysis of the application of blockchain in social emergency relief,” in *Proc. EMASS*, Wuhan, China, 2018, pp. 327–332.
- [8] J. Wang and H.-F. Chen, “Application and analysis of block chain technology in Chinese charity,” *Comput. Inf. Technol.*, vol. 25, no. 6, pp. 57–59, 2017.
- [9] Y. Qin, O. Wei, L. Wei, and H. Jiao, “Research and implementation of a charity system based on blockchain,” *Comput. Era*, no. 2, pp. 62–66, 2020.
- [10] R. M. Nor, M. M. H. Rahman, T. Rahman, and A. Abdullah, “Blockchain Sadaqa mechanism for disaster aid crowd funding,” in *Proc. 6th Int. Conf. Comput. Inform.*, 2017, pp. 400–405.
- [11] D. Jayasinghe, S. Cobourne, K. Markantonakis, R. N. Akram, and K. Mayes, “Philanthropy on the blockchain,” in *Proc. IFIP Int. Conf. Inf. Secur. Theory Pract.*, Springer, 2017, pp. 25–38.
- [12] D. D. F. Maesa, P. Mori, and L. Ricci, “Blockchain based access control,” in *Proc. IFIP Int. Conf. Distrib. Appl. Interoperable Syst.*, Springer, 2017, pp. 206–220.
- [13] Project team, “NGO donation platform — system architecture,” project architecture document, 2026.